

Statistical Learning - Classification

Assignment 1

Foundations, Representation, Optimization, and Model Selection

Due: Monday, October 19, 2026, 11:59 p.m.

100 marks

Time zone: America/Toronto (Eastern Time).

Submission. This is an individual assignment. Upload written solutions, tables, and figures to the matching questions in Crowdmark. Submit one `WatIAM_A1.zip` (replace WatIAM with your Waterloo username) to the Assignment 1 Dropbox in [Learn](#). Include the Q3, Q5, and Q6 scripts, any shared helper files, `Q6_cv_results.csv`, `Q6_predictions.csv`, and a short README with run commands, software/dependencies, and random seeds. Name each main script `Q3.py`, `Q5.py`, or `Q6.py`, or use the corresponding `.R` or `.m` extension. Keep full code in the ZIP; written answers should explain your work concisely.

Programming. Use Python, R, or MATLAB. Basic arrays, linear solves, eigen/SVD routines, random-number generation, CSV input/output, and plotting are permitted. Implement the central methods and optimization update loops yourself. Do not call ready-made PCA, Fisher discriminant analysis, LDA/QDA, logistic regression, neural-network or backpropagation, automatic differentiation, optimization, cross-validation, or early-stopping routines. You may reuse your own helper functions across questions. Use relative data paths and document numerical safeguards and convergence criteria.

Notation. Matrices are uppercase, vectors are lowercase bold, and scalars are lowercase. Observations are columns: $X = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{d \times n}$. All logarithms are natural; $\sigma(t) = 1/(1 + e^{-t})$ is the sigmoid, I_d is the identity matrix, and tr denotes the sum of diagonal entries. $\mathbf{1}\{\cdot\}$ is an indicator.

Generative AI survey (required). After completing the assignment, open [Learn](#) and choose **Submit** → **Surveys** → **Assignment 1: Generative AI Use Survey**. Complete it even if you used no generative AI (answer No to all five questions). AI use is permitted; your survey responses do not affect your assignment grade. The survey is named so completion can be checked. You remain responsible for understanding your submitted work.

Individual Verification of Understanding: You may be asked to complete a brief individual quiz or participate in an oral discussion about your assignment. This may include explaining your reasoning and solutions, answering related questions, or explaining and making small modifications to your submitted code. You must demonstrate your own understanding of all submitted work, including work developed with permitted generative AI assistance. These checks must be completed without generative AI or assistance from others. If you cannot demonstrate adequate understanding of your submitted work, your overall assignment mark may be reduced.

Late Submission Policy: Submissions received up to 24 hours after the deadline receive a 10% penalty. Submissions more than 24 hours late receive a grade of zero.

Question 1: Bayes decisions and loss functions**[12 marks]**

Let $y \in \{0, 1\}$, with priors $\pi_0, \pi_1 > 0$ and $\pi_0 + \pi_1 = 1$. Suppose

$$x \mid y = 0 \sim \mathcal{N}(\mu_0, \sigma^2), \quad x \mid y = 1 \sim \mathcal{N}(\mu_1, \sigma^2), \quad \mu_1 > \mu_0, \quad \sigma > 0.$$

A Bayes action minimizes posterior expected loss $r(a \mid x) = \sum_y \ell(a, y)P(y \mid x)$. Any minimizing action is acceptable at an equal-risk boundary.

- [3]** Under 0-1 loss $\ell_{01}(a, y) = \mathbf{1}\{a \neq y\}$, derive the Bayes classifier from the densities and priors. Write it as $h^*(x) = \mathbf{1}\{x > t\}$ and give t explicitly.
- [3]** Derive the Bayes error in closed form using the standard normal CDF Φ . Explain why it can be positive even when all distributional parameters are known exactly.
- [3]** A false positive costs $c_{fp} > 0$ and a false negative costs $c_{fn} > 0$; correct decisions cost zero. Derive the optimal posterior-probability threshold and threshold in x . Explain mathematically how the boundary changes as c_{fn} increases.
- [3]** Return to 0-1 loss and add a *reject* action costing $c_r \in (0, 1/2)$ regardless of the class. For $\eta(x) = P(y = 1 \mid x)$, derive the regions for predicting 0, rejecting, and predicting 1.

Deliverables for Question 1

Crowdmark Q1: Upload your derivations and final answers for parts (a)–(d): the Bayes classifier and threshold, Bayes-error formula and explanation, asymmetric-loss thresholds and boundary analysis, and the three reject-option decision regions.

Code: No code file is required for this question.

Question 2: LDA, QDA, and covariance geometry**[13 marks]**

Consider $\mathbf{x} \mid y = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$, $k \in \{0, 1\}$, with equal priors and

$$\boldsymbol{\mu}_0 = -\mathbf{m}, \quad \boldsymbol{\mu}_1 = \mathbf{m}, \quad \Sigma_0 = I_d, \quad \Sigma_1 = I_d + \rho \mathbf{u} \mathbf{u}^T,$$

where $\mathbf{m} \neq \mathbf{0}$, $\rho > 0$, and $\|\mathbf{u}\|_2 = 1$. You may use

$$(I_d + \rho \mathbf{u} \mathbf{u}^T)^{-1} = I_d - \frac{\rho}{1 + \rho} \mathbf{u} \mathbf{u}^T, \quad |I_d + \rho \mathbf{u} \mathbf{u}^T| = 1 + \rho.$$

- [5]** Starting from the Gaussian densities, derive the QDA discriminant difference $g(\mathbf{x}) = \delta_1(\mathbf{x}) - \delta_0(\mathbf{x})$. Obtain and simplify an explicit equation for the decision boundary $g(\mathbf{x}) = 0$.
- [3]** Prove that the decision depends only on the projection onto $\text{span}\{\mathbf{m}, \mathbf{u}\}$, the subspace generated by these two vectors. Show what happens when a vector orthogonal to both is added to $\mathbf{x}$.
- [2]** Take the limit $\rho \rightarrow 0$. Show algebraically how the boundary becomes the corresponding LDA boundary and interpret its geometry.
- [3]** A sample covariance can be singular or poorly conditioned (its smallest eigenvalues can be very small relative to its largest). For a symmetric positive-semidefinite $\hat{\Sigma}_k$ with positive trace, consider the *shrinkage* estimator

$$\hat{\Sigma}_k(\alpha) = (1 - \alpha) \hat{\Sigma}_k + \alpha \frac{\text{tr}(\hat{\Sigma}_k)}{d} I_d, \quad 0 \leq \alpha \leq 1.$$

It mixes the sample covariance with a spherical covariance of the same average variance. Describe how each eigenvalue changes with α and interpret the class covariance geometry at $\alpha = 0$ and $\alpha = 1$. The two classes may have different traces.

Deliverables for Question 2

Crowdmark Q2: Upload the QDA discriminant and boundary derivation, the projection-dependence proof, the limiting LDA boundary and geometric interpretation, and the shrinkage eigenvalue analysis and explanations for parts (a)–(d).

Code: No code file is required for this question.

Question 3: PCA versus supervised separation

[18 marks]

Let $d \geq 2$ and $y \in \{-1, +1\}$ with equal probabilities. Suppose

$$\mathbf{x} \mid y \sim \mathcal{N}(y a \mathbf{e}_2, \Sigma), \quad \Sigma = \text{diag}(v, 1, \dots, 1), \quad a > 0, \quad v > 0,$$

where $\mathbf{e}_j$ is the j th coordinate vector.

- [3] Derive the covariance of the unlabeled distribution of $\mathbf{x}$. Determine the first principal-component direction as a function of a and v , including the tied-eigenvalue case.
- [2] Find a condition on v under which PC1 has direction $\mathbf{e}_1$ (up to sign) and explains more than 95% of the total variance.
- [3] Under that condition, show that y is independent of the one-dimensional PC1 representation. Deduce the smallest classification error attainable using only PC1.
- [3] Derive the population two-class Fisher linear discriminant direction. Compute the Bayes error after projection onto it and compare with part (c).
- [7] Generate one dataset with $n = 200$, $d = 10$, $a = 2$, and $v = 300$, drawing exactly 100 independent observations from each class-conditional distribution. Choose and report a fixed random seed. Reuse this dataset for 20 randomly generated stratified 70/30 splits: split each class separately into 70 training and 30 test observations.

For each split, use these steps for both representations:

- Estimate the mean from the training observations and subtract that same mean from training and test observations. **Do not standardize feature variances in this question.**
- Estimate PC1 from the centered training data. Let $\mathcal{T}$ denote the training indices for the current split. For Fisher's direction, let $\hat{\boldsymbol{\mu}}_+$ and $\hat{\boldsymbol{\mu}}_-$ be the training class means and define

$$C_k = \frac{1}{70} \sum_{i \in \mathcal{T}: y_i = k} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^T, \quad k \in \{-1, +1\}.$$

Use a direction proportional to $(C_+ + C_-)^{-1}(\hat{\boldsymbol{\mu}}_+ - \hat{\boldsymbol{\mu}}_-)$. Normalize each estimated direction to unit length. Use basic linear algebra to implement both methods.

- For each direction $\mathbf{w}$, project using $t = \mathbf{w}^T(\mathbf{x} - \hat{\boldsymbol{\mu}})$, where $\hat{\boldsymbol{\mu}}$ is the overall training mean. Let $\hat{m}_+$ and $\hat{m}_-$ be the projected training class means. Use the supplied nearest-class-mean rule:

$$\hat{y}(t) = \begin{cases} +1, & (\hat{m}_+ - \hat{m}_-) \left(t - \frac{\hat{m}_+ + \hat{m}_-}{2} \right) > 0, \\ -1, & \text{otherwise.} \end{cases}$$

Compute the test classification error for each representation.

Report the mean and sample standard deviation (divisor 19) of the 20 test errors for each method. Include one comparison figure and explain differences from the population calculations. The repeated splits overlap; their standard deviation is a descriptive measure of split variability.

Deliverables for Question 3

Crowdmark Q3: Upload the derivations and explanations for parts (a)–(d), your random seed, the mean and sample standard deviation of test error for both methods in part (e), one comparison figure, and your discussion of the simulation results. Do not paste the full code into Crowdmark.

Code / Learn: Name your main code file `Q3.py`, `Q3.R`, or `Q3.m`. Include it and any required helper files in the single `WatIAM_A1.zip` submitted to the Assignment 1 Dropbox in Learn, together with the Q5 and Q6 files and README.

Question 4: Multiclass logistic regression

[17 marks]

This question extends binary logistic regression to k classes. Let

$$X = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{d \times n}, \quad y_i \in \{1, \dots, k\}, \quad B = [\mathbf{b}_1, \dots, \mathbf{b}_k] \in \mathbb{R}^{d \times k}.$$

Define logits $s_{ji} = \mathbf{b}_j^T \mathbf{x}_i$ and softmax probabilities

$$p_{ji} = P(y_i = j \mid \mathbf{x}_i) = \frac{e^{s_{ji}}}{\sum_{r=1}^k e^{s_{ri}}}.$$

Let $T = [t_{ji}] \in \{0, 1\}^{k \times n}$ be the one-hot label matrix, with $t_{ji} = \mathbf{1}\{y_i = j\}$, and let $P = [p_{ji}]$. Assume labels are conditionally independent given their predictors, so the likelihood is the product of the probabilities assigned to the observed labels. The likelihood *score* is the gradient of the log likelihood with respect to its parameters; it is distinct from a prediction score.

- (a) [4] Derive the log likelihood $\ell(B)$ and show that its class- j score block is

$$\nabla_{\mathbf{b}_j} \ell(B) = \sum_{i=1}^n (t_{ji} - p_{ji}) \mathbf{x}_i.$$

Give the matrix expression for $\nabla_B \ell(B)$ and verify its dimensions.

- (b) [3] Show that replacing every $\mathbf{b}_j$ by $\mathbf{b}_j + \mathbf{c}$, for any $\mathbf{c} \in \mathbb{R}^d$, leaves all probabilities unchanged. Explain why this is *non-identifiability*: distinct parameters give the same probabilities. Set the reference-class vector $\mathbf{b}_k = \mathbf{0}$ and recover binary logistic regression when $k = 2$.
- (c) [5] With $\mathbf{b}_k = \mathbf{0}$, derive the Hessian blocks for free classes $j, r \in \{1, \dots, k-1\}$:

$$H_{jr} = - \sum_{i=1}^n p_{ji} (\mathbf{1}\{j = r\} - p_{ri}) \mathbf{x}_i \mathbf{x}_i^T.$$

Show the full Hessian is negative semidefinite, meaning $\mathbf{v}^T H \mathbf{v} \leq 0$ for every compatible vector $\mathbf{v}$. Explain what concavity implies about any finite stationary point. Do not assume concavity alone ensures existence or uniqueness of a finite maximizer.

- (d) [2] *Iteratively reweighted least squares (IRLS)*. For binary $\mathbf{y} \in \{0, 1\}^n$, at a finite iterate $\boldsymbol{\beta}^{(t)}$ put $p_i^{(t)} = \sigma((\boldsymbol{\beta}^{(t)})^T \mathbf{x}_i)$ in $\mathbf{p}^{(t)}$ and define

$$W^{(t)} = \text{diag}(p_i^{(t)}(1 - p_i^{(t)})), \quad \mathbf{z}^{(t)} = X^T \boldsymbol{\beta}^{(t)} + (W^{(t)})^{-1}(\mathbf{y} - \mathbf{p}^{(t)}).$$

Assume X has full row rank in this part. The next Newton iterate minimizes

$$(\mathbf{z}^{(t)} - X^T \boldsymbol{\beta})^T W^{(t)} (\mathbf{z}^{(t)} - X^T \boldsymbol{\beta}).$$

Explain why each of the terms *iteratively*, *reweighted*, and *least squares* describes the update, and explain its connection to Newton-Raphson. A derivation of the formula is not required.

(e) [3] Use $d = 2$, $k = 3$, $n = 3$, and

$$X = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Set $\mathbf{b}_3 = \mathbf{0}$ and initialize $\mathbf{b}_1 = \mathbf{b}_2 = \mathbf{0}$. Compute the initial P and the two free score blocks. Take one gradient-ascent step with step size $\gamma = 0.3$ and compute the new probabilities for all three observations, to three decimal places.

Deliverables for Question 4

Crowdmark Q4: Upload the likelihood and score derivation with dimensions, the non-identifiability and binary-recovery arguments, the Hessian derivation and negative-semidefiniteness proof, the concavity and IRLS explanations, and all numerical results from part (e).

Code: A calculator or short script may be used to check part (e), but no Q4 code file is required.

Question 5: Backpropagation through a branched graph

[15 marks]

A *computational graph* has nodes for inputs, parameters, and intermediate quantities, with edges showing computational dependencies. A forward pass evaluates them in order. A reverse pass applies the chain rule from the loss back through the graph, summing contributions from all branches. The *adjoint* of a vector node $\mathbf{q}$ is $\bar{\mathbf{q}} = \partial\ell/\partial\mathbf{q}$; for a scalar q , it is $\bar{q} = \partial\ell/\partial q$.

For one observation $(\mathbf{x}, y)$, with $\mathbf{x} \in \mathbb{R}^d$ and $y \in \{0, 1\}$, use $W_1 \in \mathbb{R}^{r_1 \times d}$, $W_2 \in \mathbb{R}^{r_2 \times r_1}$, $\mathbf{b}_1, \mathbf{a} \in \mathbb{R}^{r_1}$, $\mathbf{b}_2, \mathbf{v} \in \mathbb{R}^{r_2}$, and $c \in \mathbb{R}$. Consider

$$\begin{aligned} \mathbf{h}_1 &= \tanh(W_1 \mathbf{x} + \mathbf{b}_1), & \mathbf{h}_2 &= \tanh(W_2 \mathbf{h}_1 + \mathbf{b}_2), \\ z &= \mathbf{v}^T \mathbf{h}_2 + \mathbf{a}^T \mathbf{h}_1 + c, & p &= \sigma(z), & \ell &= -y \log p - (1 - y) \log(1 - p). \end{aligned}$$

Here $\tanh$ acts coordinatewise and $\tanh'(t) = 1 - \tanh^2(t)$. Notice that $\mathbf{h}_1$ reaches the loss through two paths.

(a) [8] Draw the graph and derive, by a reverse pass, explicit expressions for

$$\nabla_{W_1} \ell, \quad \nabla_{\mathbf{b}_1} \ell, \quad \nabla_{W_2} \ell, \quad \nabla_{\mathbf{b}_2} \ell, \quad \nabla_{\mathbf{v}} \ell, \quad \nabla_{\mathbf{a}} \ell, \quad \frac{\partial \ell}{\partial c}.$$

Show the intermediate adjoints; final formulas alone are not sufficient.

(b) [5] Implement the forward and backward passes using basic arrays and matrix operations. Use

$$\begin{aligned} \mathbf{x} &= \begin{bmatrix} 0.6 & -1.2 & 0.3 & 0.8 \end{bmatrix}^T, & y &= 1, \\ W_1 &= \begin{bmatrix} 0.2 & -0.1 & 0.4 & 0.3 \\ -0.3 & 0.5 & -0.2 & 0.1 \\ 0.1 & 0.2 & 0.3 & -0.4 \end{bmatrix}, & \mathbf{b}_1 &= \begin{bmatrix} 0.1 \\ -0.2 \\ 0.05 \end{bmatrix}, \\ W_2 &= \begin{bmatrix} 0.3 & -0.4 & 0.2 \\ -0.1 & 0.5 & 0.3 \end{bmatrix}, & \mathbf{b}_2 &= \begin{bmatrix} 0 \\ 0.1 \end{bmatrix}, & \mathbf{v} &= \begin{bmatrix} 0.7 \\ -0.6 \end{bmatrix}, \\ \mathbf{a} &= \begin{bmatrix} 0.2 & -0.3 & 0.4 \end{bmatrix}^T, & c &= -0.1. \end{aligned}$$

Let $\boldsymbol{\theta}$ list all scalar parameter entries. For $\epsilon = 10^{-6}$ compare each backpropagation derivative with the centered finite difference

$$g_{\text{fd},j} = \frac{\ell(\boldsymbol{\theta} + \epsilon \mathbf{e}_j) - \ell(\boldsymbol{\theta} - \epsilon \mathbf{e}_j)}{2\epsilon}$$

at these nine coordinates (subscripts here are 1-based):

$$(W_1)_{11}, (W_1)_{24}, (\mathbf{b}_1)_3, (W_2)_{12}, (W_2)_{23}, (\mathbf{b}_2)_1, (\mathbf{v})_2, (\mathbf{a})_3, c.$$

Report p , ℓ , all nine gradient pairs, and the maximum scaled discrepancy over those coordinates,

$$\max_j \frac{|g_{\text{bp},j} - g_{\text{fd},j}|}{\max\{1, |g_{\text{bp},j}|, |g_{\text{fd},j}|\}}.$$

- (c) [2] The empirical risk is the average of n per-example losses. Form a mini-batch of size b by drawing indices uniformly with replacement. At fixed data and parameters, prove that its average gradient is unbiased for the full empirical-risk gradient. State the randomness over which the expectation is taken.

Deliverables for Question 5

Crowdmark Q5: Upload the computational graph, reverse-pass derivation with intermediate adjoints and all requested gradients, the numerical values of p and ℓ , a table of all nine gradient pairs, the maximum scaled discrepancy, and the mini-batch unbiasedness proof. Do not paste the full code into Crowdmark.

Code / Learn: Name your main code file `Q5.py`, `Q5.R`, or `Q5.m`. Include it and any required helper files in the same `WatIAM_A1.zip` submitted to the Assignment 1 Dropbox in Learn, together with the Q3 and Q6 files and README.

Question 6: Model selection and complexity control

[25 marks]

Dataset downloads. Extract the four CSVs listed below from the student ZIP package. The same files are embedded in this PDF: save the PDF and use the links or the Attachments panel in a PDF reader that supports attachments. Browser previews may hide attachments. Only the four named CSVs are needed for this question. Put them in the directory used by your Q6 code.

- [A1_train.csv](#): row y contains 600 labels in $\{0, 1\}$; rows $\mathbf{x}_1, \dots, \mathbf{x}_{20}$ contain the features for columns `obs001, ..., obs600`.
- [A1_test.csv](#): the same 20 feature rows for 300 unlabeled observations, `test001, ..., test300`.
- [A1_noise_train.csv](#): 100 noise-feature rows $\mathbf{z}_1, \dots, \mathbf{z}_{100}$, aligned with the labeled training observations; used in part (c).
- [A1_resampling.csv](#): the fixed `fold` row (values 1–5) for part (b) and `early_stop` row (`fit/validation`) for part (c).

In each file, the first row contains observation IDs and the first column contains row names. Match observations by ID and store predictors as $X \in \mathbb{R}^{d \times n}$. Do not use test features to fit preprocessing or choose models.

Conventions for this question. For any training subset $\mathcal{T}$ of size n_{tr} , estimate

$$\bar{x}_j = \frac{1}{n_{\text{tr}}} \sum_{i \in \mathcal{T}} x_{ji}, \quad s_j^2 = \frac{1}{n_{\text{tr}}} \sum_{i \in \mathcal{T}} (x_{ji} - \bar{x}_j)^2.$$

Transform training and held-out values by $(x_{ji} - \bar{x}_j)/s_j$ using these training-only values. For Gaussian models, use the maximum-likelihood covariance definitions below on the standardized training subset:

$$\hat{\pi}_k = \frac{n_k}{n_{\text{tr}}}, \quad \hat{\mu}_k = \frac{1}{n_k} \sum_{i \in \mathcal{T}: y_i=k} \mathbf{x}_i,$$

$$\hat{\Sigma}_k = \frac{1}{n_k} \sum_{i \in \mathcal{T}: y_i=k} (\mathbf{x}_i - \hat{\mu}_k)(\mathbf{x}_i - \hat{\mu}_k)^T, \quad \hat{\Sigma} = \sum_{k=0}^1 \hat{\pi}_k \hat{\Sigma}_k.$$

Here n_k counts training observations in class k and $\hat{\Sigma}$ is the shared LDA covariance. **Use the covariance definitions given here for this assignment.**

For held-out observations, let $p_i = P(y_i = 1 \mid \mathbf{x}_i)$ and predict 1 when $p_i \geq 0.5$. Classification error is the fraction of incorrect predictions. Log loss is the average

$$-\frac{1}{n_{\text{val}}} \sum_{i \in \mathcal{V}} [y_i \log p_i + (1 - y_i) \log(1 - p_i)].$$

Use numerically stable probability and log-loss calculations. Penalty terms are not included in validation log loss.

- (a) [4] *Model-selection autopsy*. Hyperparameters are settings, such as projection dimension and penalty strength, selected using validation data. Five-fold cross-validation fits on four folds and evaluates on the remaining fold, cycling through all five held-out folds. *Data leakage* occurs when evaluation observations influence fitted preprocessing or model fitting.

Consider this procedure: standardize the entire labeled dataset once; compute PCA once on that entire dataset; run five-fold cross-validation over dimension r and penalty λ ; choose the pair with the smallest CV error; report that same minimum as an unbiased estimate of future error. Identify the leakage and optimistic bias you can justify. Rewrite the procedure correctly, distinguishing model selection from performance estimation. Describe a valid independent performance-evaluation strategy; you do not need to implement it.

- (b) [15] *Comparative model selection*. Using only `A1_train.csv` and the fixed folds, compare the following 27 candidates:

- (i) LDA, an untuned baseline.
- (ii) Shrinkage QDA with

$$\hat{\Sigma}_k(\alpha) = (1 - \alpha)\hat{\Sigma}_k + \alpha \frac{\text{tr}(\hat{\Sigma}_k)}{d} I_d, \quad \alpha \in \{0, 0.1, 0.3, 0.5, 0.7, 0.9\}.$$

- (iii) PCA followed by L_2 -regularized binary logistic regression, using

$$r \in \{2, 5, 10, 15, 20\}, \quad \lambda \in \{0.1, 1, 10, 100\}.$$

PCA uses the standardized training features; do not rescale the principal-component scores. The case $r = 20$ retains all features through an orthogonal rotation and is equivalent to L_2 -regularized logistic regression on the standardized features. It is the baseline with no dimension reduction.

For projected vectors $\mathbf{t}_i \in \mathbb{R}^r$, use $p_i = \sigma(b + \beta^T \mathbf{t}_i)$ and maximize the **sum** log likelihood with an unpenalized intercept:

$$\ell_\lambda(b, \beta) = \sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)] - \frac{\lambda}{2} \|\beta\|_2^2.$$

Implement Newton-Raphson or a gradient-based update loop yourself. Report your convergence rule and check that it is met.

Within each fold, fit standardization, PCA, covariance estimates, and all other fitted quantities using that fold's training portion only. Apply the fitted transformations unchanged to its validation portion.

For every candidate, compute mean CV classification error and mean CV log loss, with a descriptive fold-based standard error for each: the sample standard deviation of the five fold values divided by $\sqrt{5}$. Save all 27 candidate rows in `Q6_cv_results.csv`, including model, hyperparameters, both means, and both standard errors; leave inapplicable hyperparameter cells blank. In Crowdmark, give one compact

table showing LDA and the best candidate within each tuned family, briefly compare their validation performance, and identify the overall selection.

Select the overall model and hyperparameters by lowest mean CV log loss. For an exact tie, prefer LDA, then QDA, then PCA-logistic; within QDA prefer larger α , and within PCA-logistic prefer smaller r , then larger λ . Do not describe the selected CV score as an unbiased test-error estimate.

Refit the selected pipeline, including preprocessing, on all 600 labeled observations and output the class-1 probabilities for the 300 test observations in their original order. If PCA-logistic is selected, keep the selected λ unchanged and retain the summed objective above. Under this convention, its penalty weight per observation is λ/n_{tr} ; it is therefore 20% smaller when refitting on 600 observations than when fitting a 480-observation fold. The hidden labels are used only to check pipeline consistency; a lower hidden-test error does not earn additional marks.

- (c) [6] *Early stopping as regularization.* Append the 100 noise features to the original 20. Use the fixed `early_stop` split and standardize using the `fit` subset only. Use all 120 features, an intercept, and no PCA or L_2 penalty.

Starting at $b = 0$, $\beta = \mathbf{0}$, run 3000 full-batch gradient-descent updates on the **average** negative log likelihood with step size 0.05. Full-batch means the gradient uses every fit observation; update b and β simultaneously using the same current iterate. Record fit and validation losses at iteration 0 and after every 10 completed updates through 3000. Plot both curves and choose the earliest recorded iteration attaining the lowest validation loss. Explain why choosing an earlier iterate can act as regularization, and distinguish the selected validation loss from an independent performance estimate.

Prediction file format. `Q6_predictions.csv` must have exactly two rows and 301 comma-separated cells per row, without an additional index or header row:

```
observation,test001,test002,...,test300
p1,0.12345678,0.23456789,...,0.34567890
```

The values above are illustrative; write all 300 IDs and your probabilities, not the ellipses. Probabilities must lie in $[0, 1]$. Export them with full double-precision accuracy (for example, 17 significant digits); scientific notation is accepted. Do not round them to a fixed six or eight decimal places. For any grading calculation of log loss from the submitted CSV, probabilities will be clipped to $[10^{-15}, 1 - 10^{-15}]$ before taking logarithms. This grading convention does not replace the requirement to compute validation log loss stably during model selection.

Deliverables for Question 6

Crowdmark Q6: Upload the leakage and model-selection analysis from (a); the compact three-family comparison table, selected model and hyperparameters, convergence rule and checks, and performance discussion from (b); and the fit/validation-loss figure, selected stopping iteration, and explanations from (c). Do not paste the full code or all 300 test probabilities into Crowdmark.

Code / Learn: Name your main code file `Q6.py`, `Q6.R`, or `Q6.m`. Include it, `Q6_cv_results.csv` (all 27 candidates), `Q6_predictions.csv` (the two-row format above), and any required helper files in `WatIAM_A1.zip`, together with the Q3 and Q5 code and README. Submit this **one ZIP file** to the Assignment 1 Dropbox in Learn.